

Multiple Choice (10 marks)

1. Explain how an estimate of the age of a rock is made on the basis of the radioactive elements present.
 - (a) By assessing the fossil content found within the rock layers
 - (b) By comparing the density of the rock to a known age database
 - (c) By examining the color and texture of the rock
 - (d) By counting the layers of the rock
 - (e) By measuring the amount of carbon-14 remaining in the rock
2. What are some of the cues used by animal in migrating?
 - (a) Animals use the sound of the ocean waves for directional cues in migration.
 - (b) Animals rely solely on memory from previous migrations.
 - (c) Animals migrate based on the patterns of the stars alone.
 - (d) Animals follow changes in air pressure for migration.
 - (e) Animals use temperature changes for migration.
3. A DNA molecule has 180 base pairs and is 20 percent adenine. How many cytosine nucleotides are present in this molecule of DNA?
 - (a) 108 nucleotides
 - (b) 120 nucleotides
 - (c) 150 nucleotides
 - (d) 144 nucleotides
 - (e) 60 nucleotides
4. Synthesis of an RNA/DNA hybrid from a single-stranded RNA template requires
 - (a) RNase only
 - (b) a DNA primer and RNase
 - (c) a DNA or RNA primer and DNA polymerase
 - (d) a DNA or RNA primer and reverse transcriptase
 - (e) a RNA primer and DNA polymerase
5. How do bacteria develop drug resistance?
 - (a) Antibiotic resistance arises from bacteria absorbing nutrients that counteract the drug

- (b) Resistance develops due to prolonged exposure to antibiotics
- (c) Bacteria develop a natural immunity to drugs over time
- (d) Bacteria develop resistance through spontaneous generation without exposure to antibiotics
- (e) Drug resistance is caused solely by the use of antibacterial hand sanitizers

True / False (10 marks)

Answer True or False

- 6. True or False: Are hepatitis G virus and TT virus involved in cryptogenic chronic liver disease?
- 7. True or False: Is there any relationship between streptococcal infection and multiple sclerosis?
- 8. True or False: Can medical students contribute to quality assurance programmes in day surgery?
- 9. True or False: Does pain intensity predict a poor opioid response in cancer patients?
- 10. True or False: Eyelid-parotid metastasis: do we screen for coexisting masses?

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

- 11. Discuss the role of low external pH in ATP production by isolated mitochondria, specifically analyzing how it affects the diffusion of H^+ ions between the intermembrane space and the matrix.
- 12. Discuss the role of fermentation in cellular respiration when oxygen is unavailable, focusing on how it regenerates NAD^+ and its implications for energy production in organisms.

End of Paper